ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

NATURAL SCIENCES

PRACTICE EXAM 1 · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 8 Natural Sciences practice exam 1.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words if the science is correct. Accept other correct examples. Penalise missing units once per question.

SECTION A

QUESTION 1

1 1.1 B ✓ 1.2 D ✓ 1.3 B ✓ 1.4 C ✓ 1.5 A ✓ 1.6 B ✓ 1.7 D ✓ 1.8 A ✓ 1.9 C ✓ 1.10 C ✓ (10)

QUESTION 2

2 2.1 Chlorophyll ✓ 2.2 Food web ✓ 2.3 Ecosystem ✓ 2.4 Micro-organisms ✓ 2.5 Compound ✓ 2.6 Diffusion ✓ 2.7 Density ✓ 2.8 Series circuit ✓ 2.9 Refraction ✓ 2.10 Light-year ✓ (10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3

3.1 carbon dioxide + water ✓ —(light energy, chlorophyll)→ ✓ glucose + oxygen ✓ (3)

3.2.1 To remove the chlorophyll (green colour) so that the colour change can be seen ✓ (1)

3.2.2 Blue-black ✓ (1)

3.2.3 No starch will be found (iodine stays brown/orange) ✓; without light the plant could not photosynthesise, and it used up its stored starch ✓. (2)

3.3.1 Grass ✓ (1)

3.3.2 Any TWO: rabbit ✓; mouse ✓; grasshopper (2)

3.3.3 Any correct chain of four, e.g. grass → grasshopper → frog → snake ✓✓ (or grass → mouse → snake → eagle) (2)

3.3.4 The number of frogs would increase ✓ because snakes (their predators) would no longer eat them ✓. (2)

3.3.5 The Sun ✓ (1)

3.4 (a) Any TWO: yeast is used to make bread ✓; bacteria are used to make yoghurt, cheese or amasi ✓; decomposers recycle nutrients; some are used to make medicines (antibiotics) and vaccines; bacteria in our intestines help digestion (4)
(b) A vaccine contains weakened or dead germs (or parts of them) ✓; the body makes antibodies against them, so it can fight the real germ quickly in future (immunity) ✓.

3.5 Any ONE: photosynthesis makes glucose and stores energy, respiration breaks down glucose and releases energy; photosynthesis uses carbon dioxide and gives off oxygen, respiration does the opposite; photosynthesis happens only in light in green plants, respiration happens all the time in all living cells ✓ (1)

[20]

QUESTION 4

4.1 (a) Solid: particles packed closely together in a regular pattern ✓; they vibrate in fixed positions ✓.
(b) Gas: particles very far apart, with no pattern ✓; they move fast and freely in all directions ✓. (4)

4.2 The hot water heats the metal lid, so its particles move faster and further apart (it expands) ✓. Metal expands more than glass, so the lid becomes looser ✓. (2)

4.3 Perfume particles evaporate and move randomly among the air particles ✓; they spread from where there are many of them to where there are few (diffusion) until they reach your nose ✓. (2)

4.4 $\text{(a) density} = 540 \div 200 \checkmark = 2{,}7\ \text{g/cm}^3 \checkmark$ (b) Sink – it is denser than water ✓ (3)

4.5 (a) Element ✓ (b) Compound ✓ (c) Element ✓ (d) Compound ✓ (4)

4.6 2 hydrogen, 1 sulfur and 4 oxygen atoms ✓✓ (7 atoms in total) (2)

4.7 (a) Iron and oxygen ✓ (b) Iron oxide (rust) ✓ (c) Any ONE: a new substance with different properties forms; colour change (grey to reddish-brown) ✓ (3)

[20]

QUESTION 5

5.1 Electrons are transferred from the hair to the balloon ✓. The balloon becomes negatively charged and the hair positively charged ✓. Opposite charges attract, so the hair is pulled towards the balloon (and the hairs, all positive, repel each other and stand up) ✓. (3)

5.2 Any TWO: go indoors or into a car ✓; stay away from tall trees, poles and open fields ✓; do not swim; do not use a landline telephone or touch metal taps; crouch low if caught outside (2)

5.3.1 Circuit B ✓ (1)

5.3.2 Circuit B ✓ (1)

5.3.3 Circuit A: the other bulb goes out because the only path is broken ✓. Circuit B: the other bulb keeps glowing (just as brightly) because it is on its own path ✓. (2)

5.3.4 Parallel ✓; each light can be switched on and off separately, and if one bulb blows the others still work ✓. (2)

5.4 (a) The current decreases ✓ (b) Any ONE: kettle, iron, toaster, heater, stove ✓ (2)

5.5.1 Red, orange, yellow, green, blue, indigo, violet ✓✓ (one mark if one colour is out of place) (2)

5.5.2 The tomato reflects red light ✓ and absorbs all the other colours ✓. (2)

5.5.3 The angle of incidence is equal to the angle of reflection ✓ (1)

5.5.4 Refraction ✓: light bends (changes direction) as it passes from water into air, so the straw appears bent ✓ (2)

[20]

QUESTION 6

6.1 Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune ✓✓✓✓ (one mark for every two planets in the correct position) (4)

6.2 Any TWO: inner planets are small and rocky, outer planets are large gas (and ice) giants ✓; inner planets have few or no moons, outer planets have many moons ✓; outer planets have rings; inner planets are closer to the Sun and warmer (2)

6.3 A dwarf planet ✓, because it has not cleared other objects out of its orbit ✓ (it is small and shares its orbit with many icy objects). (2)

6.4 Between Mars and Jupiter ✓ (1)

6.5 A ball of ice, frozen gases and dust ✓; its tail always points away from the Sun ✓. (2)

6.6 (a) The Milky Way ✓ (b) A spiral (barred spiral) ✓ (2)

6.7 About 4,2 years ✓ (1)

6.8 Any TWO: communication (phone, internet, TV) ✓; weather forecasting ✓; GPS navigation; studying the Earth (mapping, farming, disasters); research in space (2)

6.9 (a) The Northern Cape (Karoo) ✓ (b) Radio waves ✓ (2)

6.10 Neil Armstrong ✓ in 1969 ✓ (2)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF MARKING GUIDELINE